TASK - 1

AI FUNDAMENTALS AND PROMPT ENGINEERING

INTERNSHIP:GROWFINIX AI INTERNSHIP

SUBMITTED BY : DHARSHINI S

COLLEGE: AMRITA VISHWA VIDYAPEETHAM

DEPARTMENT : ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

TASK:AI FUNDAMENTALS AND PROMPT ENGINEERING

DATE:19 OCTOBER 2026

EXPERIMENT -1

PROMPT 1:BASIC PROMPT

Explain artificial intelligence to a first-year college student.

1. Give a simple definition.
2. Explain three real-world applications.
3. Give one example from college life.
4. Use headings and bullet points.
5. Use simple English.

AI OUTPUT:

Artificial Intelligence (AI)

1. Simple Definition

Artificial Intelligence (AI) is a type of technology that allows computers and machines to perform tasks that usually require human intelligence.

- AI can learn from information.
- It can solve problems and answer questions.
- It can understand language and recognize images.
- It can help people make decisions.

In simple words, AI makes computers smart enough to perform tasks that normally need human thinking.

2. Three Real-World Applications of AI

A. Healthcare

AI helps doctors find diseases and provide better treatment.

- It can help detect diseases from medical scans.
- It can study patient information to support doctors.
- It can help hospitals manage patient records.

Example: AI can help a doctor identify signs of cancer in an X-ray or other medical scan.

B. Transportation

AI helps make transportation safer and more efficient.

- It helps self-driving cars recognize roads and obstacles.
- It helps navigation apps suggest faster routes.
- It helps predict traffic conditions.

Example: Google Maps uses AI-based methods to estimate travel times and suggest routes.

C. Education

AI helps students and teachers learn and teach more effectively.

- It explains difficult topics in simple language.
- It helps students practice questions and improve their skills.
- It can help teachers prepare learning materials.

Example: AI chatbots can explain a difficult mathematics problem step by step.

3. One Example from College Life

Imagine you are a first-year college student preparing for an exam.

You can use an AI tool like ChatGPT to help you study.

- Understanding topics: Ask AI to explain a difficult concept in simple English.
- Preparing notes: Ask it to create short notes from your study material.
- Practicing for exams: Ask it to generate practice questions and quizzes.
- Improving writing: Ask it to check grammar in your assignment.

Example: You ask ChatGPT, “Explain photosynthesis in simple English.” It gives you an easy explanation that helps you understand the topic.

However, you should check important facts and follow your college's rules for using AI. AI can sometimes give incorrect answers, so it should support your learning rather than replace your own thinking.

Conclusion:

- Artificial Intelligence is technology that helps machines perform tasks that usually require human intelligence.
- AI is used in healthcare, transportation, education, and many other fields.
- In college, AI can help students understand lessons, prepare for exams, and improve their skills.

Remember: AI is a tool that can make learning and everyday tasks easier when used wisely.

ergsy.com
picory.com

EXPERIMENT-2

IMPROVED PROMPT:

Act as an AI professor teaching an AI and Data Science student.

Explain Artificial Intelligence in simple English.

Include:
1. A definition in two sentences.
2. Three real-world applications with examples.
3. One example from college life.
4. A comparison table of traditional programming and AI.
5. A five-question quiz with answers at the end.

Keep the explanation beginner-friendly and well organized.

AI OUTPUT:

Artificial Intelligence (AI): A Beginner's Guide

Welcome to your first AI lecture! As an AI and Data Science student, you will learn how computers use data to recognize patterns, solve problems, and make predictions.

1. Definition of Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that usually require human intelligence. These tasks include learning from data, understanding language, recognizing images, solving problems, and making decisions.

In simple words, AI helps computers perform tasks that normally need human thinking.

2. Three Real-World Applications of AI

Application 1: Healthcare

How AI is used:

- Helps doctors identify signs of diseases.
- Analyzes medical images, such as X-rays.
- Helps predict health risks using patient data.

Example: An AI system examines a chest X-ray and highlights areas that a doctor should check carefully.

Application 2: Transportation

How AI is used:

- Helps self-driving cars recognize roads and obstacles.
- Predicts traffic conditions.
- Suggests suitable routes based on traffic data.

Example: A navigation app suggests a different route when it detects heavy traffic.

Application 3: Banking and Finance

How AI is used:

- Detects unusual banking transactions.
- Helps banks assess loan applications.
- Supports customer service through chatbots.

Example: If your bank card is suddenly used for an unusual transaction, an AI system may flag it for further checking.

3. An Example from College Life

Imagine that you have an assignment on Machine Learning, but you find the topic difficult to understand.

You use an AI tool such as ChatGPT to help you study.

- Learning: Ask AI to explain machine learning in simple English.
- Data analysis: Use an AI tool to identify patterns in a dataset for a college project.
- Exam preparation: Ask AI to create practice questions from your notes.
- Programming: Ask AI to explain an error in your Python code.

Example prompt:

> Explain machine learning to a first-year college student and give two simple examples.

AI can provide an explanation and examples to help you learn. However, always verify important information and make sure you understand the work you submit.

4. Traditional Programming vs. Artificial Intelligence

Both traditional programming and AI help computers solve problems, but they work in different ways.

Feature	Traditional Programming	Artificial Intelligence
Basic idea	Follows rules written by a programmer	Can learn patterns from data in many AI systems
Input	Data and programmed rules	Data and, during training, often examples with expected answers
How it works	Executes predefined instructions	Uses a trained model to produce predictions or outputs

Learning	Does not learn automatically from new data	Some AI systems can be trained or updated using new data
Best suited for	Tasks with clear rules	Tasks involving patterns, language, images, or predictions
Example	Calculating a student's total marks	Predicting whether a student may need extra academic support

Simple example:

Suppose you want to identify spam emails.

- Traditional programming: A programmer writes rules such as “mark an email as spam if it contains certain suspicious words.”
- AI: A model learns patterns from many examples of spam and non-spam emails and uses those patterns to classify new emails.

Note: Traditional programs can also use data, and AI systems still rely on programmed instructions. The key difference is that machine-learning systems learn patterns from examples instead of relying only on manually written decision rules.

5. Five-Question Quiz

Test your understanding of the lesson with these five questions.

What is Artificial Intelligence (AI)?

1 of 5

Next

Final Takeaway

As an AI and Data Science student, remember these three points:

- AI enables computers to perform tasks that usually require human intelligence.
- Data helps many AI systems discover patterns and make predictions.

- Machine learning (ML) is a branch of AI in which systems learn patterns from data.

These are the foundations you will build on when studying Python, statistics, data analysis, and machine learning.

COMPARISON BETWEEN PROMPT 1 AND PROMPT 2

Prompt 1: gives a general explanation.
Prompt 2: requests specific examples, a comparison table and a quiz.
Adding context and clear instructions helps make the output more suitable for the intended purpose.

EXPERIMENT-3

TEXT SUMMARIZATION USING AI

Summarize the following paragraph into exactly 5 bullet points. Use simple English and preserve the most important information.

Paragraph:

Artificial Intelligence is a branch of computer science that enables machines to perform tasks that usually require human intelligence. These tasks include learning, reasoning, problem-solving, language understanding, and recognizing images. AI is widely used in healthcare to assist doctors, in education to personalize learning, in banking to detect fraud, and in transportation to support navigation. Machine learning is a subset of AI that allows systems to learn patterns from data and improve their performance. Although AI offers many benefits, it also creates challenges related to privacy, bias, security, and employment.

Definition: Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that usually require human intelligence.

Main abilities: AI can learn, reason, solve problems, understand language, and recognize images.

Real-world uses: AI is used in healthcare, education, banking, and transportation.

Machine learning: It is a part of AI that learns patterns from data and improves performance.

Benefits and challenges: AI offers many benefits but also raises concerns about privacy, bias, security, and employment.

OBSERVATION:

The AI tool summarized the given paragraph into five concise bullet points while retaining the main ideas. The summary was easy to understand and saved time compared with reading the entire paragraph. However, the generated summary should be checked for accuracy and missing information.

EXPERIMENT-4

CONTENT GENERATION USING AI

PROMPT:
Act as a professional communication assistant.

Write a polite and professional email from a college student to a project guide requesting guidance for an AI and Data Science project.

Requirements:
1. Include a clear subject line.
2. Introduce the student briefly.
3. Explain the purpose of the email.
4. Request a convenient time for a discussion.
5. Thank the professor politely.
6. Keep the email between 100 and 150 words.
7. Use a formal tone and proper email format.

AI OUTPUT :

Dear Professor,

I hope you are doing well.

My name is Dharshini S, and I am a student pursuing 2nd year in Artificial Intelligence and Data Science at Amrita Vishwa Vidyapeetham.

I am writing to request your guidance for my upcoming AI and Data Science project. I would appreciate your advice on selecting a suitable project topic, defining the objectives, and identifying the appropriate tools and technologies to use. Your guidance would help me plan the project effectively and improve my technical knowledge.

Could you please let me know a convenient time when I may meet with you to discuss the project? I will be happy to adjust my schedule according to your availability.

Thank you for your valuable time and consideration. I look forward to learning from your guidance.

Yours sincerely,
Dharshini S
2nd year Aids
Amrita Vishwa Vidyapeetham

Observation:

The AI generated a professional email with a clear subject line, polite language, and a structured format. Specific instructions helped produce content suitable for academic communication.

EXPERIMENT- 5

INFORMATION EXTRACTION USING AI :

Act as a data extraction assistant.

Read the paragraph below and extract the following information into a table:

- Student name

- College

- Project title

- Programming language

- Project deadline

If any information is missing, write "Not provided". Do not invent information.

Paragraph:

Priya is a second-year student at ABC Engineering College. She is developing an AI-based Student Performance Prediction project using Python. Her project guide has asked her to submit the project by 15 November 2026.

Information	Details
Student Name	Priya
College	ABC Engineering College
Project Title	AI-based Student Performance Prediction
Programming Language	Python
Project Deadline	15 November 2026

OBSERVATION:

The AI extracted specific details from unstructured text and organized them into a structured table. The instruction not to invent missing information helps reduce unsupported answers.

CONCLUSION:

Through this task, I explored the fundamentals of Artificial Intelligence and learned how prompt engineering can improve the quality and usefulness of AI-generated responses. I experimented with basic and improved prompts, text summarization, content generation, and information extraction.

These experiments helped me understand the importance of clear instructions, relevant context, structured output formats, and verification of AI-generated information. I also learned that AI tools can support academic learning, professional communication, and data organization.

Overall, this task strengthened my understanding of practical AI applications and provided a foundation for further learning in Python programming, AI model integration, and application development.